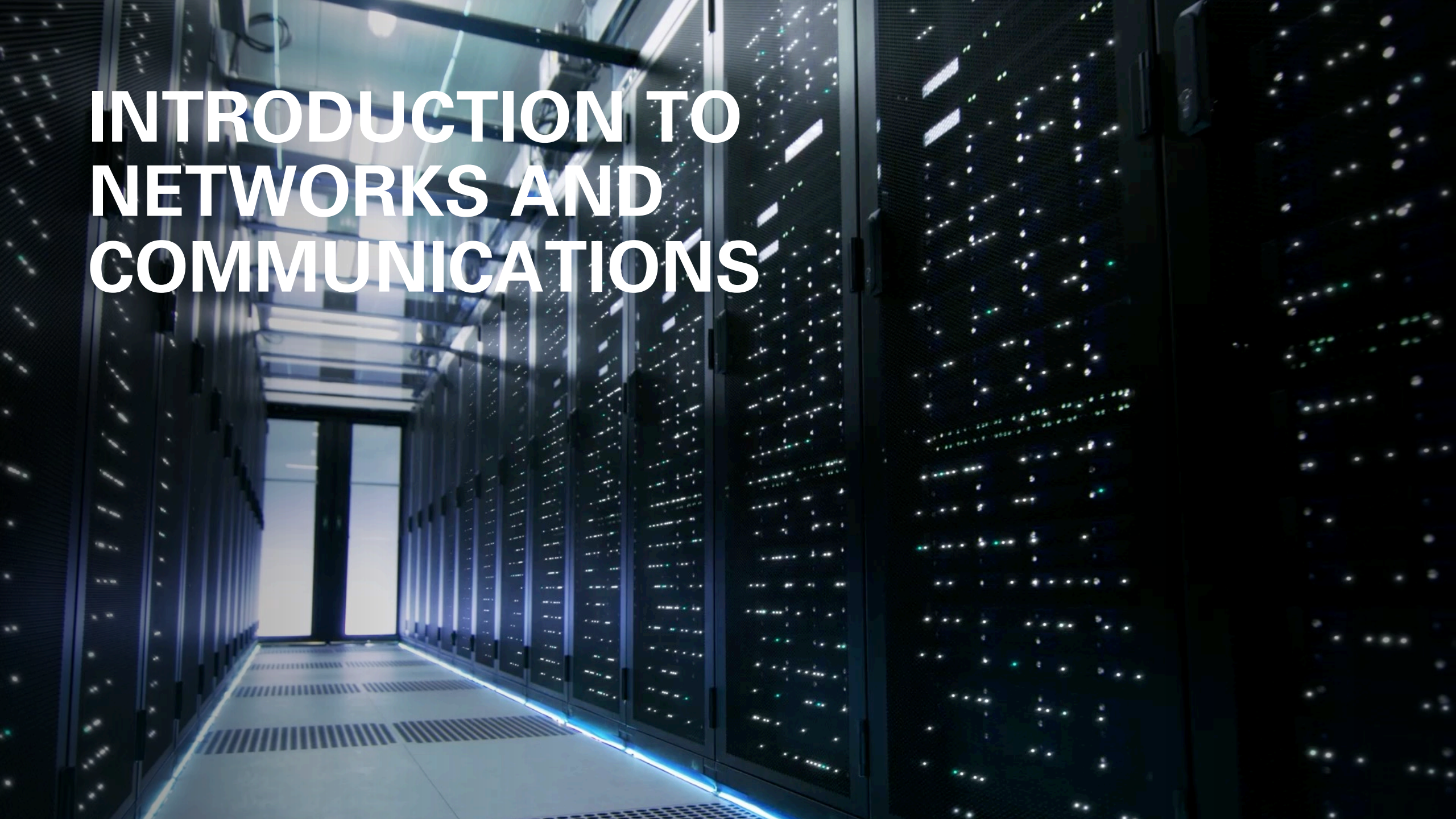


INTRODUCTION TO NETWORKS AND COMMUNICATIONS



Objectives:

- Define computer networks
- Discuss data communications and its components.
- Differentiate the types of data flow.
- Identify the Network Criteria
- Define each network topology and how it works.

Introduction to Computer Networks



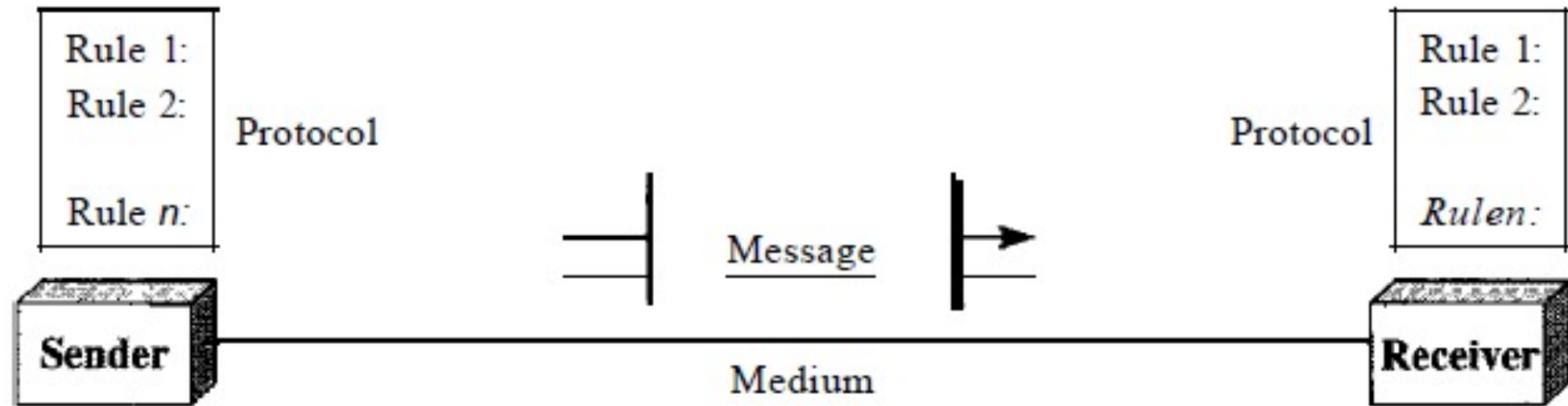
What is a Computer Network?

- Computer network, two or more computers that are connected with one another for the purpose of communicating data electronically.
- Two basic network types are local-area networks (LANs) and wide-area networks (WANs).

What is Data Communications?

- Data communications are the exchange of data between two devices via some form of transmission medium such as a wire cable.
- For data communications to occur, the communicating devices must be part of a communication system made up of a combination of hardware (physical equipment) and software (programs).

Components of Data Communications



Components of Data Communications

1. Message

- The **message** is the information (data) to be communicated. Popular forms of information include text, numbers, pictures, audio, and video.

2. Sender

- The sender is the device that sends the data message. It can be a computer,
- workstation, telephone handset, video camera, and so on.

Components of Data Communications

3. Receiver.

The receiver is the device that receives the message. It can be a computer, workstation, telephone handset, television, and so on.

4. Transmission medium.

The transmission medium is the physical path by which a message travels from sender to receiver.

Some examples of transmission media include twisted-pair wire, coaxial cable, fiber-optic cable, and radio waves.

Components of Data Communications

- 5. Protocol.
 - A protocol is a set of rules that govern data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating, just as a person speaking French cannot be understood by a person who speaks only Japanese.

An Introduction to Data Commu nications



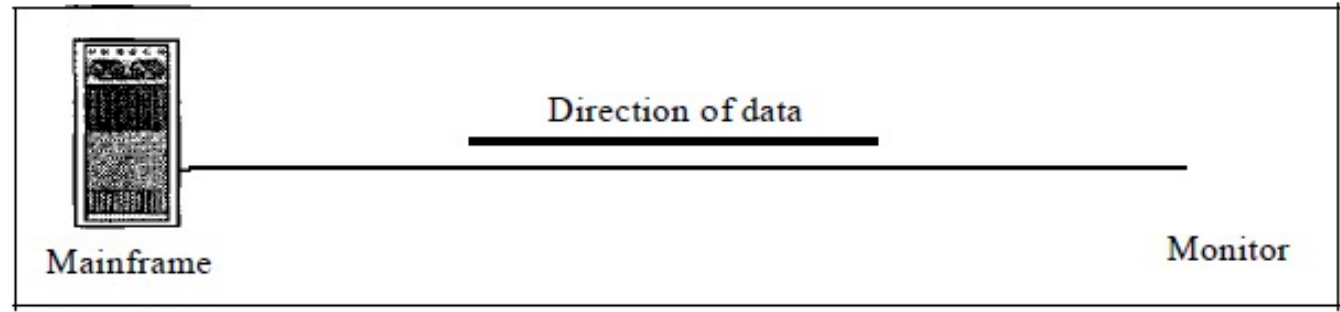
Data Representation

- Text
- Numbers
- Images
- Audio
- Video

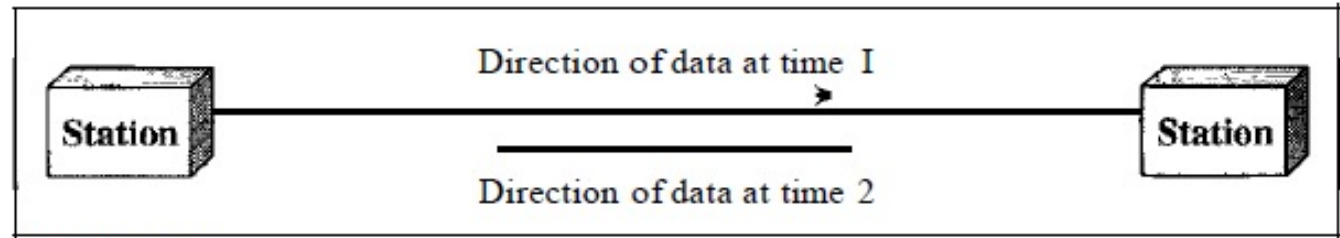
Data Flow

Communication between two devices can be

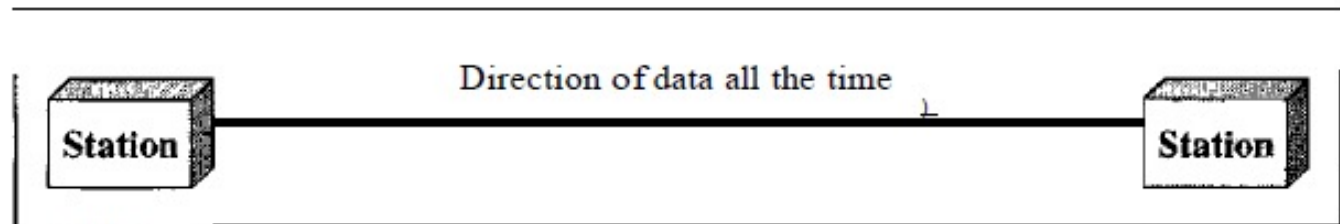
- simplex,
- half-duplex, or
- full-duplex.



a. Simplex



b. Half-duplex



c. Full-duplex

Data Flow-Simplex

- In **simplex mode**, the communication is unidirectional, as on a one-way street. Only one of the two devices on a link can transmit; the other can only receive.
- Keyboards and traditional monitors are examples of simplex devices. The keyboard can only introduce input; the monitor can only accept output. The simplex mode can use the entire capacity of the channel to send data in one direction.

Data Flow-Half-Duplex

- In half-duplex mode, each station can both transmit and receive, but not at the same time: When one device is sending, the other can only receive, and vice versa.
- The half-duplex mode is like a one-lane road with traffic allowed in both directions.
- When cars are traveling in one direction, cars going the other way must wait.
- In a half-duplex transmission, the entire capacity of a channel is taken over by whichever of the two devices is transmitting at the time.

Data Flow-Half-Duplex

- Walkie-talkies and CB (citizens band) radios are both half-duplex systems.
- The half-duplex mode is used in cases where there is no need for communication in both directions at the same time; the entire capacity of the channel can be utilized for each direction.

Data Flow- Full-Duplex

- In full-duplex mode (also called duplex), both stations can transmit and receive simultaneously.
- The full-duplex mode is like a two-way street with traffic flowing in both directions at the same time.
- In full-duplex mode, signals going in one direction share the capacity of the link: with signals going in the other direction.

Data Flow- Full-Duplex

- This sharing can occur in two ways:
- Either the link must contain two physically separate transmission paths, one for sending and the other for receiving; or the capacity of the channel is divided between signals traveling in both directions.
- One common example of full-duplex communication is the telephone network.
- When two people are communicating by a telephone line, both can talk and listen at the same time.

Half Duplex vs. Full Duplex



Network Criteria

- A network must be able to meet a certain number of criteria. These are:
 - Performance
 - Reliability
 - Security

Performance

- Performance can be measured in many ways, including transit time and response time.
- Transit time is the amount of time required for a message to travel from one device to another.
- Response time is the elapsed time between an inquiry and a response.

Factors Affecting Network Performance



The number of users



the type of transmission medium



the capabilities of the connected hardware, and



the efficiency of the software.



Reliability

- In addition to accuracy of delivery, network reliability is measured by the frequency of failure, the time it takes a link to recover from a failure, and the network's robustness in a catastrophe.



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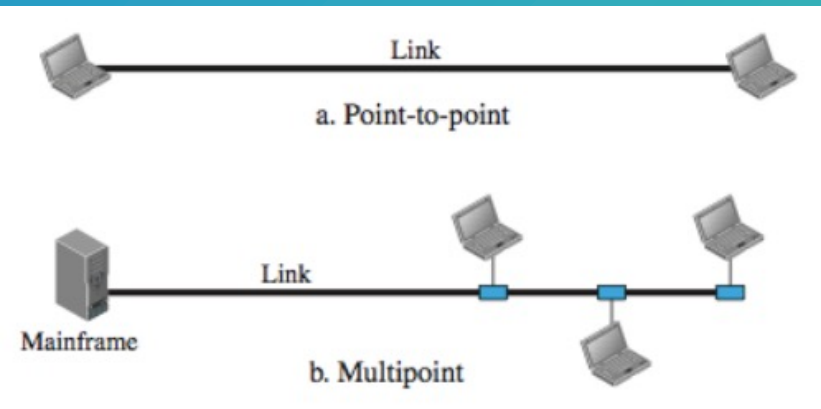
Security

- Network security issues include protecting data from unauthorized access, protecting data from damage and development, and implementing policies and procedures for recovery from breaches and data losses.

Physical Structures

- A **network** is two or more devices connected through links.
- A **link** is a communications pathway that transfers data from one device to another.
- For visualization purposes, it is simplest to imagine any link as a line drawn between two points.
- For communication to occur, two devices must be connected in some way to the same link at the same time.

Types of Connections



- Point-to-point
- Multi-point

Point-to-Point

- A **point-to-point** connection provides a dedicated link between two devices.
- The entire capacity of the link is reserved for transmission between those two devices.
- Most point-to-point connections use an actual length of wire or cable to connect the two ends, but other options, such as microwave or satellite links, are also possible.

Example of Point-to-Point

- When you change television channels by infrared remote control, you are establishing a point-to-point connection between the remote control and the television's control system.

Multipoint

- A multipoint (also called multidrop) connection is one in which more than two specific devices share a single link.
- In a multipoint environment, the capacity of the channel is shared, either spatially or temporally.
- If several devices can use the link simultaneously, it is a *spatially shared* connection.
- If users must take turns, it is a *timeshared connection*

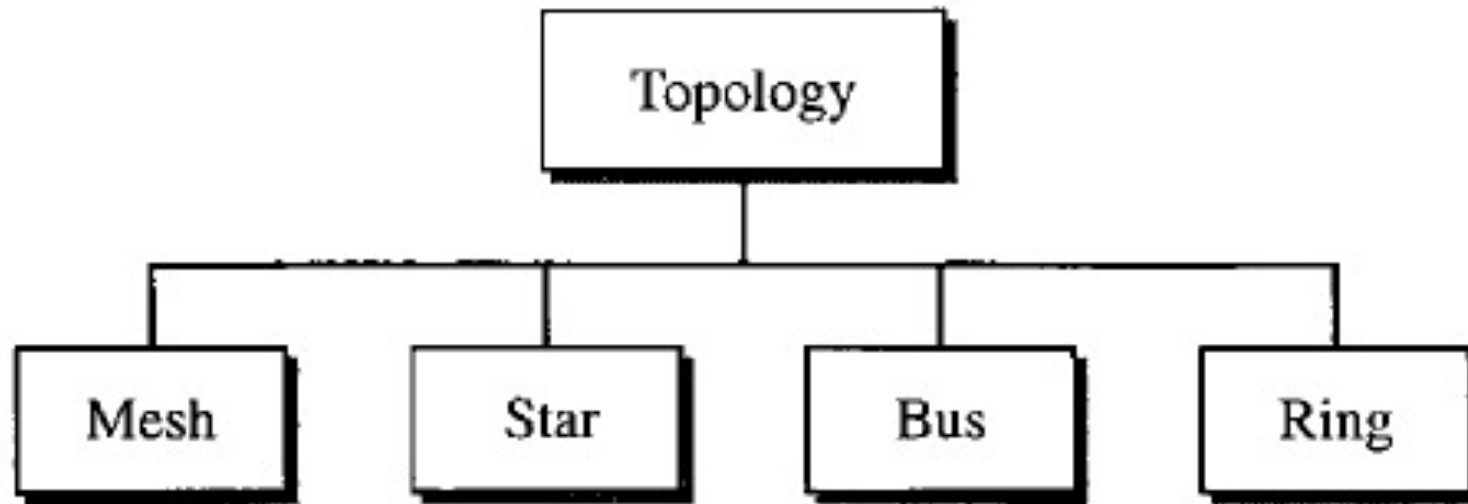
Physical Topology

- The term physical topology refers to the way in which a network is laid out physically:
 - Two or more devices connect to a link;
 - two or more links form a topology.
- The topology of a network is the geometric representation of the relationship of all the links and linking devices (usually called nodes) to one another.

Network Topology



Physical Topology





Mesh

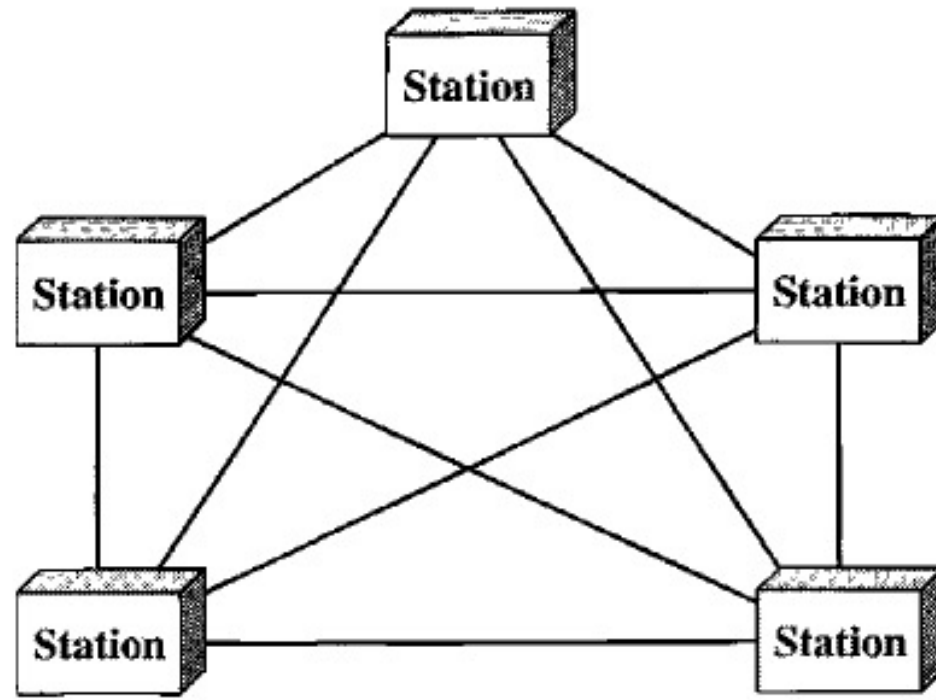
- Mesh In a mesh topology, every device has a dedicated point-to-point link to every other device.
- The term *dedicated means that the link carries traffic only between the* two devices it connects.

Mesh

- To find the number of physical links in a fully connected mesh network with n nodes, we first consider that each node must be connected to every other node.
- Node 1 must be connected to $n - 1$ nodes, node 2 must be connected to $n - 1$ nodes, and finally node n must be connected to $n - 1$ nodes. We need $n(n - 1)$ physical links.

Fully Connected Mesh

A fully connected mesh topology (five devices)



A mesh offers several advantages over other network topologies.

- First, the use of dedicated links guarantees that each connection can carry its own data load, thus eliminating the traffic problems that can occur when links must be shared by multiple devices.
- Physical boundaries prevent other users from gaining access to messages.

A mesh offers several advantages over other network topologies.

- Finally, point-to-point links make fault identification and fault isolation easy.
- Traffic can be routed to avoid links with suspected problems.
- This facility enables the network manager to discover the precise location of the fault and aids in finding its cause and solution.

Mesh-Disadvantage

- The main disadvantages of a mesh are related to the amount of cabling and the number of I/O ports required.
- First, because every device must be connected to every other device, installation and reconnection are difficult.
- Second, the sheer bulk of the wiring can be greater than the available space (in walls, ceilings, or floors) can accommodate.
- Finally, the hardware required to connect each link (I/O ports and cable) can be prohibitively expensive.

The background of the image is a dark blue gradient. Overlaid on this is a complex, abstract network of white lines and dots. The dots represent nodes, and the lines represent connections between them. The network is dense and irregular, with many small triangles and polygons formed by the connections. The overall shape of the network is elongated and somewhat wavy, spanning most of the width of the image. The text 'STAR TOPOLOGY' is centered horizontally and partially overlaps the network structure.

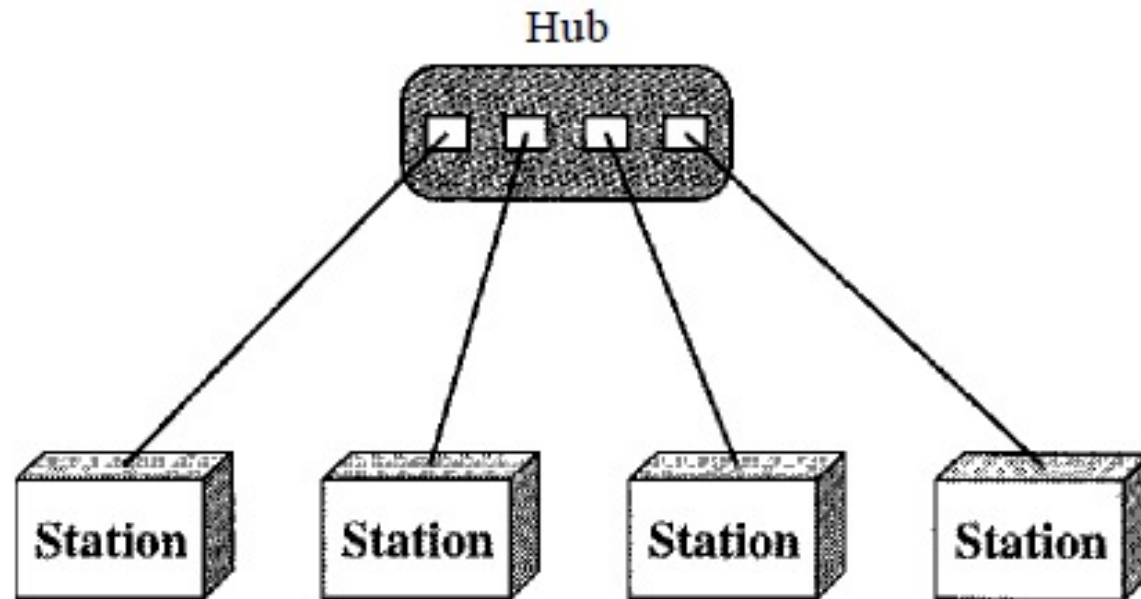
STAR TOPOLOGY

Star Topology

- In a star topology, each device has a dedicated point-to-point link only to a central controller, usually called a hub. The devices are not directly linked to one another.
- Unlike a mesh topology, a star topology does not allow direct traffic between devices.
- The controller acts as an exchange: If one device wants to send data to another, it sends the data to the controller, which then relays the data to the other connected device.

Star Topology

A star topology connecting four stations



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Advantages

- A star topology is less expensive than a mesh topology.
- In a star, each device needs only one link and one I/O port to connect it to any number of others.
- This factor also makes it easy to install and reconfigure.
- Far less cabling needs to be housed, and additions, moves, and deletions involve only one connection: between that device and the hub.

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Advantages

- Other advantages include robustness. If one link fails, only that link is affected.
- All other links remain active. This factor also lends itself to easy fault identification and fault isolation. As long as the hub is working, it can be used to monitor link problems and bypass defective links.

Disadvantage of Star Topology

- One big disadvantage of a star topology is the dependency of the whole topology on one single point, the hub. If the hub goes down, the whole system is dead.
- Although a star requires far less cable than a mesh, each node must be linked to a central hub. For this reason, often more cabling is required in a star than in some other topologies (such as ring or bus).

Applications of Star Topology



The star topology is used in local-area networks (LANs).

High-speed LANs often use a star topology with a central hub.

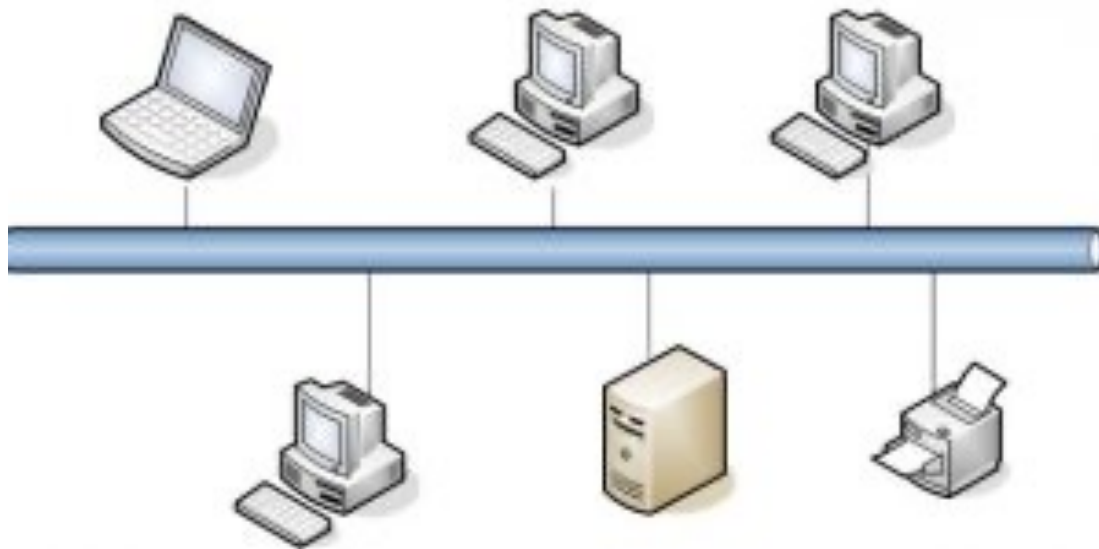


BUS TOPOLOGY

Bus Topology

- A bus topology, on the other hand, is multipoint. One long cable acts as a backbone to link all the devices in a network.
- A bus topology is used in a LAN (Local Area Network) where different nodes are linked to a particular cable otherwise a single backbone.

Bus Topology



- A coaxial cable/RJ-45 cable is used to connect different devices. In a bus topology, when the cable has any problem then the entire network will not work. For the safety purpose of a network, there may be alternate cables.

How it works

- Nodes are connected to the bus cable by drop lines and taps. A drop line is a connection running between the device and the main cable.
- A tap is a connector that either splices into the main cable or punctures the sheathing of a cable to create a contact with the metallic core.

How it works

- As a signal travels along the backbone, some of its energy is transformed into heat.
- Therefore, it becomes weaker and weaker as it travels farther and farther.
- For this reason, there is a limit on the number of taps a bus can support and on the distance between those taps.

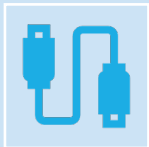


RING TOPOLOGY

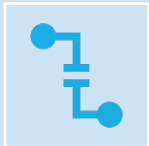
Ring Topology



In a ring topology, each device has a dedicated point-to-point connection with only the two devices on either side of it.



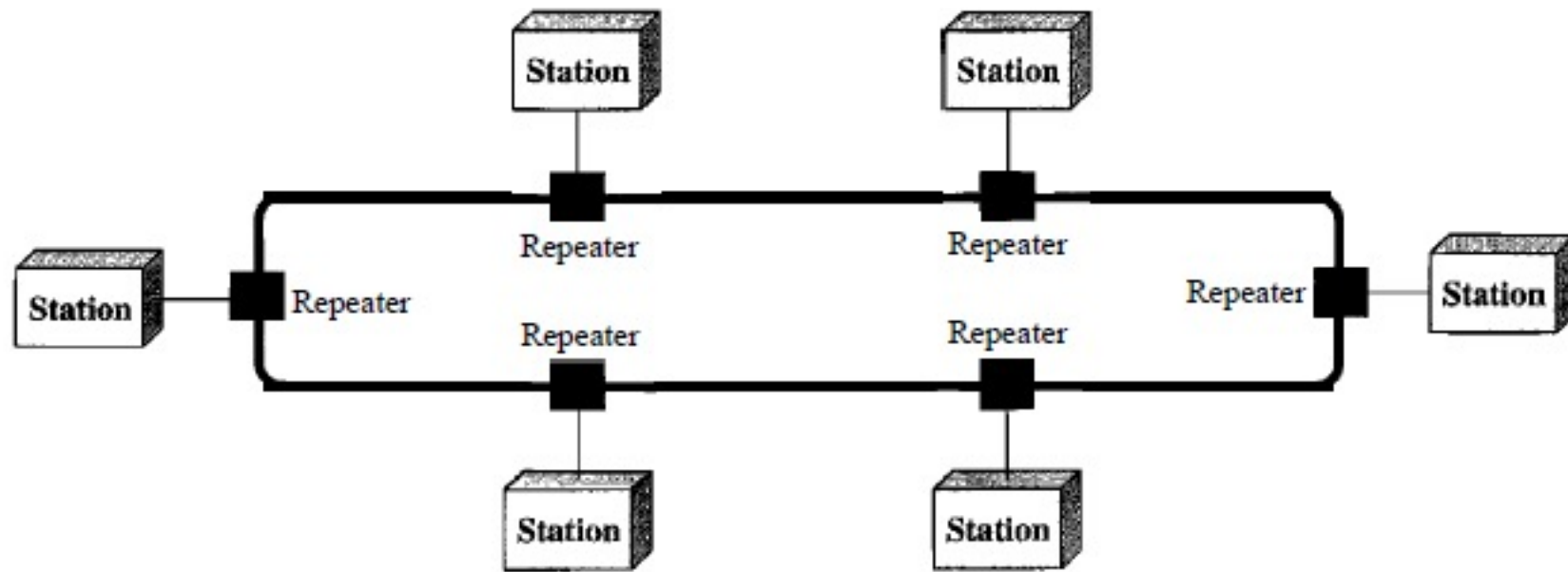
A signal is passed along the ring in one direction, from device to device, until it reaches its destination. Each device in the ring incorporates a repeater.



When a device receives a signal intended for another device, its repeater regenerates the bits and passes them along.

Ring Topology

Figure 1.8 *A ring topology connecting six stations*



Ring Topology

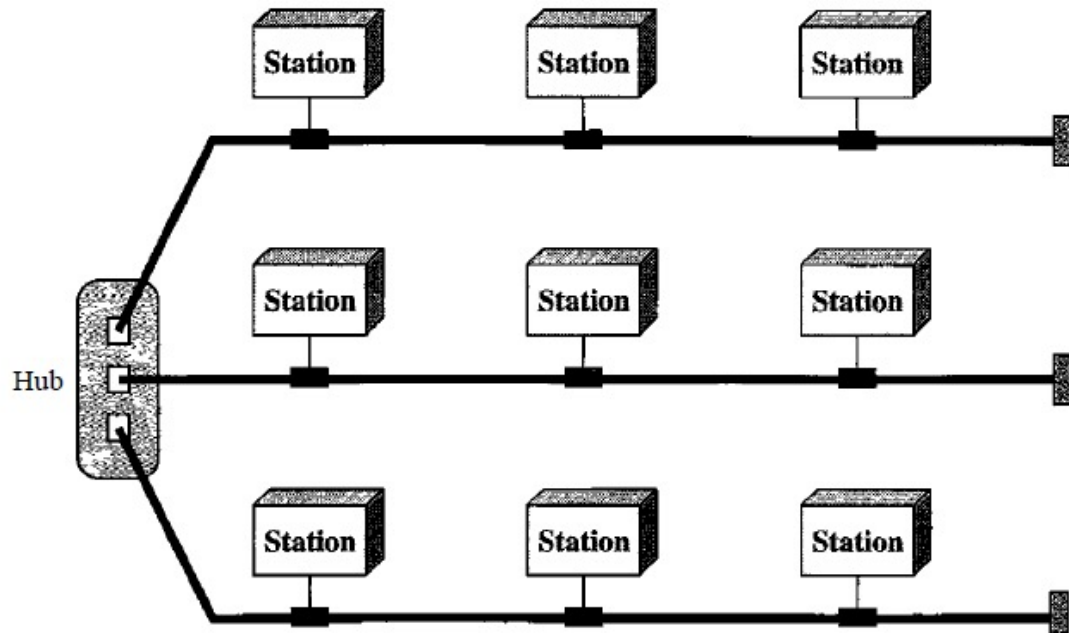
- A ring is relatively easy to install and reconfigure.
- Each device is linked to only its immediate neighbors (either physically or logically).
- To add or delete a device requires changing only two connections.

Ring Topology

- The only constraints are media and traffic considerations (maximum ring length and number of devices). In addition, fault isolation is simplified.
- Generally, in a ring, a signal is always circulating. If one device does not receive a signal within a specified period, it can issue an alarm.
- The alarm alerts the network operator to the problem and its location.

Hybrid Topology

A hybrid topology: a star backbone with three bus networks



- A network can be hybrid. For example, we can have a main star topology with each branch connecting several stations in a bus topology.

LANs and WANs

- LANs connect computers and peripheral devices in a limited physical area, such as a business office, laboratory, or college campus, by means of links (wires, Ethernet cables, fibre optics, Wi-Fi) that transmit data rapidly.
- A typical LAN consists of two or more personal computers, printers, and high-capacity disk-storage devices called file servers, which enable each computer on the network to access a common set of files.